Heuristic Usability Evaluation
OOPP Group 1

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1 INTRODUCTION

The objective of this evaluation is to find usability problems in the user interface design of our OOPP team's project, using the methods described by Jakob Nielsen in article [3], in order to address and solve each issue in a subsequent debriefing session.

The prototype under evaluation consists of an application intended for task management (similar to Trello), built using Spring-Boot, JavaFX and JPA technologies. The prototype represents a work-in-progress version of the final product that our team will deliver for the examination of this course. In the following, the evaluated state of the application will be presented in a detailed manner, with the aim of allowing the reader to replicate the evaluation. The following features were successfully implemented at the time of the assessment:

- Joining a server;
- Disconnecting and joining another server;
- Creating or joining an already existing board;
- Creating one or more lists;
- Creating one or more cards;
- Viewing and deleting said cards.

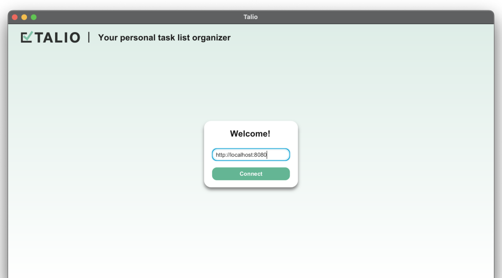

Figure 1: The main screen

Figure 11 illustrates the main screen of our application, which contains a welcome message and a text box where a pre-generated server address is found. Underneath is placed a button to "connect" to the server, which further opens a pop-up to join or create a desired board, as seen in figure 2. It is worth mentioning that upon a failed server connection (potentially caused by writing an inexistent server address in the field or trying to connect to a closed server), the alert in figure 3 pops up instead, informing the user about the situation and providing solutions to the problem.

After a successful connection to the server and input of the desired board key, the user is redirected to the workspace (figure 4), which presents the joined/created board. The currently implemented options in the interface are the ones for adding a new list by clicking the corresponding button in the upper-right corner of the workspace, deleting the shown board, creating/joining another board and ultimately disconnecting from the server.

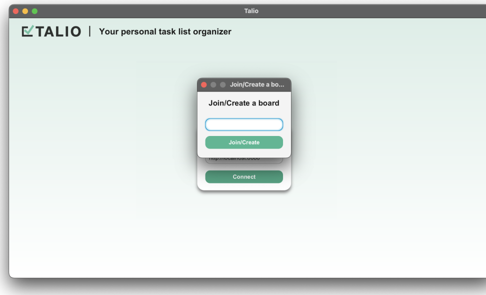

Figure 2: Join or create a board

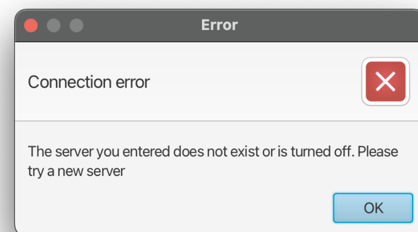

Figure 3: Error alert upon failed connection

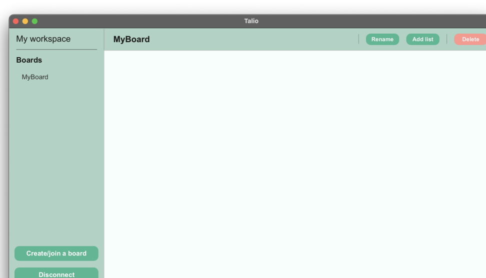

Figure 4: The workspace

The button for creating a new list opens a new pop-up which prompts the user to input a name for the new list (see figure 5), and afterwards the workspace looks like in figure 6.

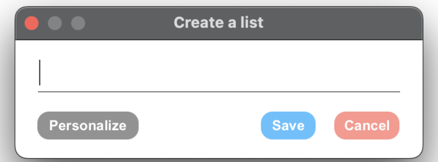

Figure 5: Creating a new list

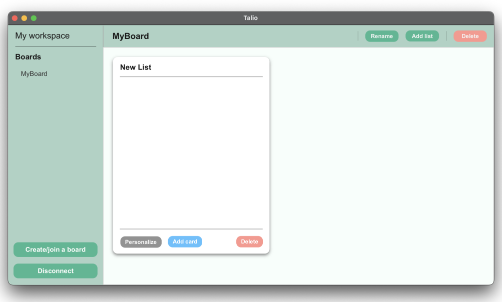

Figure 6: A new list added to the workspace

It is now possible to click the blue "Add card" button in the newly-created list to add tasks (cards) to it. This opens a pop-up designed specifically for this purpose, which you may analyze in figure 7. After inputting the desired title (and description), it is possible to save the card by pressing the corresponding button, which leads to the state depicted in figure 8.

Lastly, it is possible to view a card upon double-clicking it in the corresponding list, which then pops up the view portrayed in figure 9, which currently is only capable of being closed when pressing the button in the title bar.

2 METHODS

In this section, we will describe the methodology used when executing the evaluation.

2.1 Experts

For the purpose of the evaluation, we recruited 5 experts from OOPP Group 23. As Computer Science and Engineering students who have used other scheduling apps extensively, they are experienced with software such as Talio as both users and professionals, therefore being a perfect choice for our evaluation.

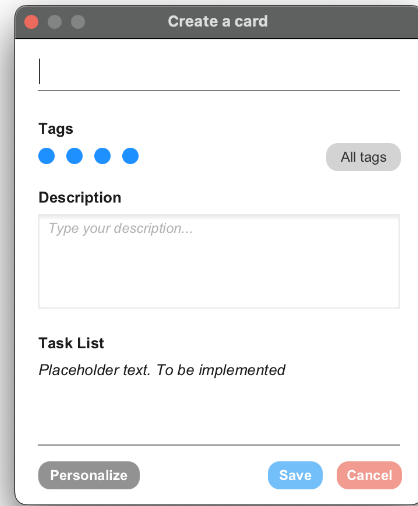

Figure 7: A new list added to the workspace

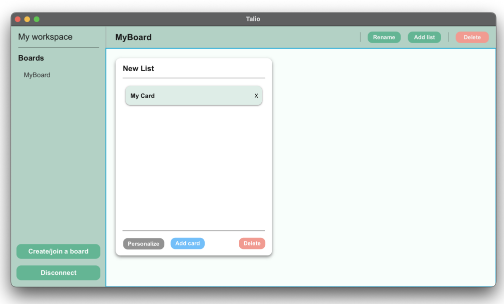

Figure 8: A new list added to the workspace

2.2 Procedure

The evaluation took place in an open environment, more specifically, in the Drebbelweg building, at TU Delft. We conformed to the standards of conducting a heuristic usability evaluation presented during our lecture and also in article [3]. The evaluators were presented with the aforementioned prototype of our application, having the features listed in the introduction of the report. Each expert received a laptop from a member of Group 1, all running the same version of the prototype app. The examiners had to assess the app in isolation.

The evaluators were allowed to explore the app freely for a period of 30 minutes. They were encouraged to use the prototype as they wished, in spite of the fact that it is not a finished product, in order to test the quality of the interface even if the functionality of certain aspects was not fully implemented.

The examiners were required to follow the 10 heuristics illustrated in [2]:

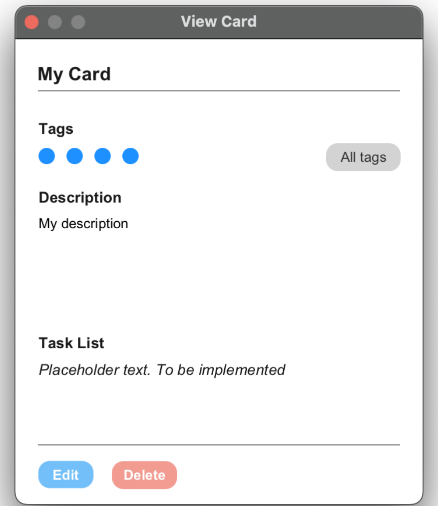

Figure 9: A new list added to the workspace

(1) Visibility of system status: Keep users informed about what is happening through appropriate feedback within a reasonable amount of time.
(2) Match between system and the real world: Use language, concepts, and metaphors that are familiar to the user and that match the user's expectations.
(3) User control and freedom: Provide users with a way to undo actions or to leave a state without penalty.
(4) Consistency and standards: Follow established conventions and user expectations to create a predictable experience.
(5) Error prevention: Design the system to prevent errors from occurring in the first place, or at least minimize their impact.
(6) Recognition rather than recall: Make objects, actions, and options visible and easily accessible to the user, rather than requiring them to remember specific details.
(7) Flexibility and efficiency of use: Provide shortcuts, accelerators, and other means for users to streamline their interactions with the system.
(8) Aesthetic and minimalist design: Strive for simplicity and elegance in the design of the user interface.
(9) Help users recognize, diagnose, and recover from errors: Provide clear, concise error messages and guidance on how to resolve them.
(10) Help and documentation: Provide users with documentation and other resources to help them learn how to use the system effectively.

2.3 Measures (Data collection)

We measured the problems arising from the design of our prototype, collected in the format presented in [1], which we also used for our raw results:

(1) Problem description: a brief description of the problem;
(2) Likely/actual difficulties: the anticipated difficulties that the user will encounter as a consequence of this problem;
(3) Specific contexts: the specific context in which the problem may occur;
(4) Assumed causes: description of the cause(s) of the problem.

The evaluators were asked to report the problems they found in the above format, while also mentioning which heuristic of the 10 previously mentioned the problem refers to, in order to allow us to measure the scale of each problem.

3 RESULTS

In this section, we will list and analyze the identified problems, which we will also prioritize using a severity matrix. Since this report is only about design, from our raw results we filtered out the problems related to functionality; we also took false positives out of consideration (for example problems about the lack of features not required by the official backlog or specifically ruled out by course staff). Additionally, we fixed small grammar and spelling issues and split the problems which were mistakenly grouped as only one into smaller, separate problems. We'll now present our results:

Problem 1

Identified Heuristic: Flexibility and efficiency of use

(1) Problem description: The buttons are not descriptive enough;
(2) Likely/actual difficulties: It might take a user a bit of time to find specific controls, for example, to rename a list;
(3) Specific contexts: When a user tries to rename a list, they presumably have to click the "personalize" button at the bottom of the list, which is counter-intuitive and they might instead click the "rename" button, which presumably renames the board;
(4) Assumed causes: Not enough consideration for the intuitiveness of controls.

This is indeed the case, as seen in figure 6. Even more, there are multiple "Delete" buttons on the screen, one in the upper-right corner (intended to delete the board), and one in every added list (intended to delete the corresponding list). As a result of the identical design, it is therefore unclear which button provides which functionality.

Problem 2

Identified Heuristic: Flexibility and efficiency of use

(1) Problem description: Board name is persistent in the text field of join/create board;
(2) Likely/actual difficulties: User might not want to connect to the same board;
(3) Specific contexts: When the user wants to join/create a board;

(4) Assumed causes: Did not clear the text field content when it is loaded.

This is also a good catch. In our design, after first inputting the board key to the board join pop-up (seen in figure 2), every time another board join pop-up is opened the preceding key is maintained in the text field, forcing the user to first clear the text and then write the key of the new board they want to join.

Problem 3

Identified Heuristic: Aesthetic and minimalist design

(1) Problem description: Pop up create a separate window which might mislead;
(2) Likely/actual difficulties: User might not like working with too many windows;
(3) Specific contexts: While creating new board, card, task, tag, etc.;
(4) Assumed causes: Unspecified.

This is a generally good idea we missed in our design: indeed, many of our pop-up windows can be removed, to conform to the minimal and aesthetic design standards in our heuristics. For example, the pop-ups in figures 2 and 5 could prove to be very annoying considering the user is forced to resolve their corresponding prompt before being able to return to the workspace. However, other pop-ups need to be kept, as an in-window design would be very cluttered, hence contradicting heuristic 7.

Problem 4

Identified Heuristic: Consistency and standards

(1) Problem description: Positioning of buttons in pop-ups is not consistent;
(2) Likely/actual difficulties: The user will keep getting confused about where to click;
(3) Specific contexts: The position of "Edit" and "Delete" in the "Card" pop-up is on the left, whereas in all the other pop-ups they are on the right;
(4) Assumed causes: Consistency was not kept in mind while designing.

This problem is seen in figure 7 and, for example, figure 5. Definitely, this can be misleading for the user and consistency should be taken into account.

Problem 5

Identified Heuristic: Flexibility and efficiency of use

(1) Problem description: Resizing the app doesn't handle full screen well despite offering the option;
(2) Likely/actual difficulties: The user can find it annoying having to scroll while at the same time, there is unused space underneath lists that could have been used and would have meant that less scrolling would be required;
(3) Specific contexts: When having many added cards, and resizing to full screen;
(4) Assumed causes: No adjustments for full-screen mode made

The causes for this problem are visible even in non-full-screen windows of the app, such as that in figure 6: more specifically, the lists in the workspace are not taking all the vertical space they can, and they remain of constant height even when the window is maximized, therefore showing only a limited amount of cards and forcing unnecessary scrolling through them.

Problem 6

Identified Heuristic: Help and documentation

(1) Problem description: In the main screen there is not much info regarding the input of the field, unless you try to input some random text and the alert pops up, which lets you know that the a server needs to be provided;
(2) Likely/actual difficulties: The user can get confused if they are not familiar with the app and its purpose;
(3) Specific contexts: Usage of the main screen of the app by a user who had not used it before;
(4) Assumed causes: The lack of instructions on the main page. The app feels minimal, which is nice, but there is not enough info provided.

Very clearly explained, this problem is one that is seen throughout the app: in figure 11 for instance, unless the text field is empty, it is impossible to know what the prompt actually is asking (i.e., inputting a server address). The same problems go when creating lists (figure 5) and cards (figure 7), where the fact that the title fields are selected by default makes it unclear what should be typed in there.

At the end of this section, we describe a prioritization of these problems illustrated in the severity matrix visible in figure 10. The main factor in deciding the imminence of a problem was its frequency and its impact, ordered using the corresponding heuristic (in our case, we considered consistency to be the most impactful issue, then flexibility, design, and lastly help and documentation).

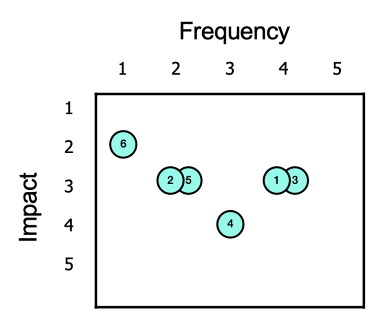

Figure 10: Severity matrix of the found problems

4 CONCLUSIONS & IMPROVEMENTS

As a general conclusion, our user interface proved to lack a sense of consistency, flexibility, and also directions related to how exactly the application works for new users. Taking the results into consideration, we prioritized solving the problems and improving the overall design. We will now discuss our final design, explaining why we applied certain changes compared to our initial prototype.

As seen in figure 11, we have redesigned the main screen to define more clearly what is expected as input ("Please input your desired server address"). This way we ensure that the user understands what input is necessary even if they are not familiar with our application, addressing the issue raised in Problem 6. The new design is in conformity with heuristic 10, as we are now providing explicit help to users.

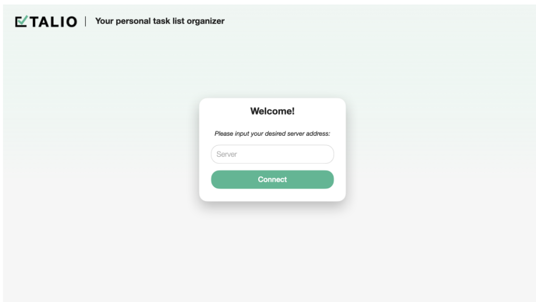

Figure 11: New main screen

Additionally, we removed the "join/create board" pop-up, as suggested by the evaluators in problem 3. Now clicking the "Connect" button redirects immediately to a blank workspace, which can be seen in figure 12. The pop-up was replaced with an in-window text field on the bottom left side of the screen that allows users to create or join other boards more efficiently, and according to the flexibility and efficiency of use standards of heuristic 7. Categorized in the same heuristic, problem 2 is also solved by the same design modification, as now the "Add" button clears the text field where the user provides a board key. Likewise, a new button for creating lists has been added, and the "Delete" and "Rename" buttons associated with boards have been renamed accordingly in order to improve clarity, solving problem 1. In the same figure, it is visible how now lists take all the vertical space of the window (which also expands in full screen), making it less likely that a user has to scroll through a card list unless at least 8 cards are added. This also addresses, problem 5, further granting flexibility to our design, as required by heuristic 7.

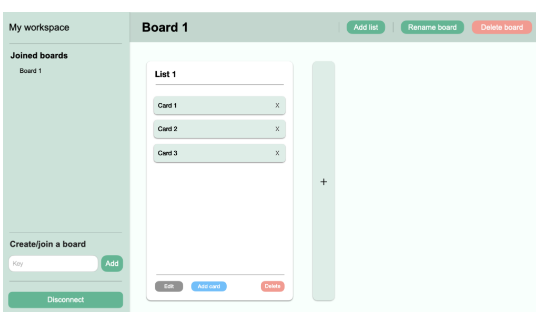

Figure 12: Improved Board and List management

Regarding the complaints raised in Problem 4, we have redesigned the "View Card" screen (figure 13): now, the "Edit" and "Delete" buttons are on the right side of the pop-up, making it consistent with the rest of the app and adhering to the principles of heuristic 4.

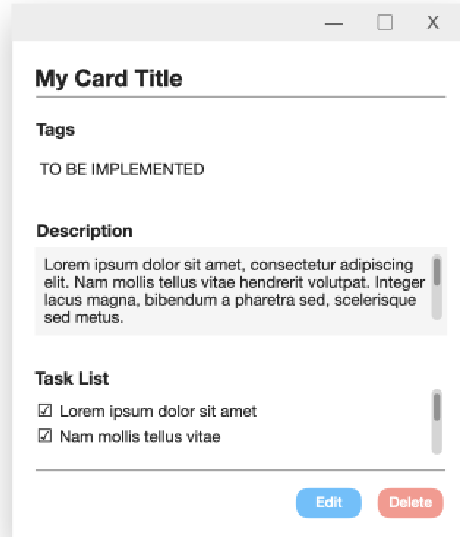

Figure 13: Updated "View Card" screen

Moreover, figure 14 illustrates how the "Add Card" screen has been improved by adding new buttons intended for task and tag creation. This improvement makes the interface more intuitive and allows the user to handle these features in an easier manner, building on the requirements of heuristics 7 and 8.

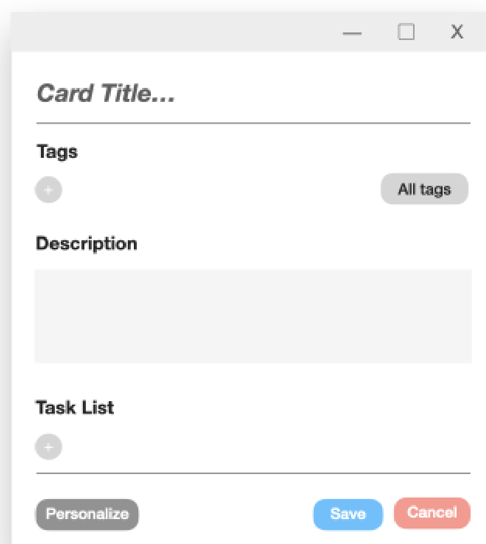

Figure 14: Updated "Add Card" Screen

Taking everything into account, our evaluation has revealed numerous issues that escaped internal testing, allowing us to later add a wide range of improvements to the overall design that is in

conformity with our heuristics. We thank the members of OOPP Team 23 for assessing our prototype in a professional way, offering us helpful feedback that proved to be both instructive and constructive.

REFERENCES

[1] Gilbert Cockton, Alan Woolrych, Lynne Hall, and Mark Hindmarch. 2004. *Changing Analysts' Tunes: The Surprising Impact of a New Instrument for Usability Inspection Method Assessment.* 145–161. https://doi.org/10.1007/978-1-4471-3754-2_9
[2] Jakob Nielsen. 1994. *Heuristic Evaluation.* John Wiley & Sons, 25–62.
[3] Jakob Nielsen. 1994. How to Conduct a Heuristic Evaluation. https://www.nngroup.com/articles/how-to-conduct-a-heuristic-evaluation. Accessed: 2023-24-03.